

JOHNSTOWN CASTLE, Ireland

WMO: 039560

Lat: 52.2978N

Lon: 6.4967W

Elev: 53

StdP: 100.69

Time Zone: 1.00 (IRL)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.7	0.3	-4.1	2.7	0.7	-2.8	3.0	1.7	13.2	8.0	11.8	7.6	3.6	340	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	20.6	16.4	19.3	15.8	18.4	15.3	17.4	19.1	16.8	18.4	16.2	17.6	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.0	17.9	16.1	11.5	17.3	15.6	11.1	16.8	49.0	19.1	47.3	18.5	45.6	17.6	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.5	8.5	-2.4	22.9	1.4	0.9	-3.4	23.6	-4.2	24.1	-5.0	24.6	-6.0	25.3		
(5)			-3.1	18.7	1.5	0.7	-4.2	19.2	-5.1	19.5	-5.9	19.9	-6.9	20.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	6.1	5.9	6.6	8.5	11.1	13.7	15.4	15.0	13.5	11.3	8.1	6.6
	DBStd	4.08	2.32	2.36	2.25	2.04	2.11	1.97	1.78	1.45	1.73	2.21	2.70	2.86
	HDD10.0	596	121	115	107	53	11	1	0	0	1	13	68	108
	HDD18.3	2979	379	348	363	295	223	140	92	104	146	219	308	363
	CDD10.0	660	1	1	2	7	46	111	168	154	105	52	10	3
	CDD18.3	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.2	4.9	4.7	4.4	4.2	4.1	3.8	3.7	3.9	3.8	4.2	4.4	4.9
Precipitation	PrecAvg	977	100	76	63	63	65	66	68	77	76	112	106	103
	PrecMax	1334	223	188	126	125	171	193	157	267	158	243	230	278
	PrecMin	749	24	19	14	19	4	9	12	5	15	25	31	30
	PrecStd	140	42	41	32	32	35	43	38	57	38	52	54	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	12.0	13.8	16.2	19.3	21.9	22.7	20.8	19.3	16.7	14.2	12.6
		MCWB	10.7	10.2	10.1	12.1	15.1	16.4	17.5	16.7	15.7	14.9	13.3	12.0
	2%	DB	11.0	10.8	12.0	14.1	17.2	19.6	21.0	19.2	18.0	15.8	13.5	12.0
		MCWB	10.4	9.6	9.3	10.8	14.0	15.5	16.8	15.9	15.1	14.5	12.8	11.3
	5%	DB	10.4	10.2	10.8	12.9	15.7	18.1	19.8	18.4	17.2	15.2	12.9	11.5
		MCWB	9.8	9.3	8.9	10.2	12.9	14.8	16.2	15.4	14.7	14.0	12.1	10.8
	10%	DB	9.9	9.6	10.0	11.7	14.6	17.1	18.7	17.7	16.4	14.6	12.3	10.8
		MCWB	9.1	8.8	8.5	9.4	12.0	14.1	15.7	15.1	14.4	13.5	11.4	10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.9	11.1	12.7	15.7	17.3	18.6	17.7	17.2	15.7	13.8	12.4
		MCDB	11.3	11.3	12.2	15.6	18.6	20.5	21.0	19.0	18.1	16.2	13.9	12.5
	2%	WB	10.7	10.2	10.3	11.5	14.5	16.1	17.7	17.1	16.2	15.1	13.0	11.7
		MCDB	10.8	10.6	11.1	13.5	16.5	18.5	19.8	18.1	17.0	15.6	13.3	11.9
	5%	WB	10.0	9.6	9.6	10.7	13.5	15.4	17.0	16.6	15.6	14.4	12.3	11.0
		MCDB	10.3	10.1	10.4	12.1	15.3	17.4	19.0	17.6	16.4	14.9	12.8	11.4
	10%	WB	9.4	9.0	8.9	10.1	12.7	14.7	16.3	16.0	15.1	13.8	11.5	10.4
		MCDB	9.8	9.6	9.8	11.3	14.2	16.5	18.1	17.1	16.0	14.4	12.2	10.9
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.7	5.5	5.7	5.7	5.8	5.7	5.6	5.5	5.0	4.7	4.5
		MCDBR	4.6	4.7	6.1	7.0	6.6	7.0	6.9	6.1	6.0	4.8	4.5	4.6
		MCWBR	4.8	4.0	4.6	4.7	4.4	4.5	4.1	3.9	4.2	4.1	4.5	5.0
	5% WB	MCDBR	4.8	4.3	5.6	6.0	6.2	6.3	6.0	5.1	4.8	4.3	4.4	4.3
		MCWBR	5.1	4.0	4.8	4.3	4.3	4.4	4.1	3.8	4.1	4.3	4.6	4.8
Clear-Sky Solar Irradiance	taub	0.333	0.344	0.371	0.381	0.389	0.376	0.383	0.363	0.359	0.350	0.327	0.321	
	taud	2.306	2.366	2.317	2.323	2.343	2.409	2.395	2.462	2.469	2.440	2.379	2.284	
	Ebn at Noon	646	765	814	853	863	877	864	864	825	750	655	585	
	Edh at Noon	64	80	102	114	117	111	111	98	87	74	59	56	
All-Sky Solar Radiation	RadAvg	0.76	1.46	2.58	4.03	5.04	5.40	4.95	4.19	3.07	1.75	0.95	0.58	
	RadStd	0.07	0.11	0.27	0.36	0.32	0.46	0.43	0.27	0.21	0.16	0.09	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page